

BridgeFinder: Finding People Across Structural Holes

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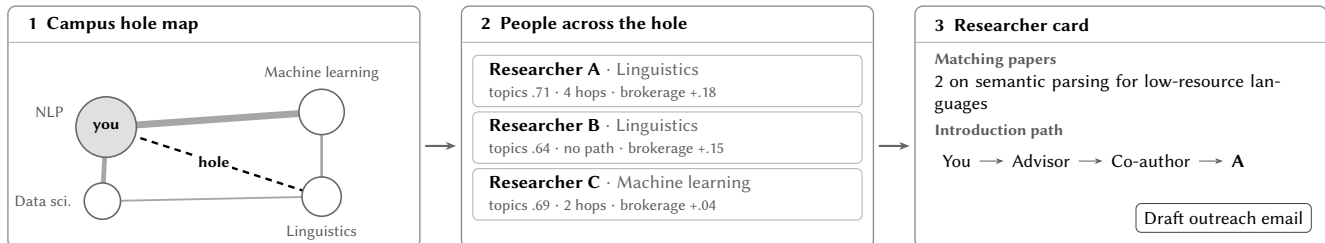


Figure 1: Core flow. (1) UIUC co-authorship communities; dashed = a structural hole (similar topics, almost no joint papers). (2) Researchers across the hole, ranked. (3) Their matching papers and an introduction path.

1 OBJECTIVE

Students seeking a research mentor, and researchers seeking collaborators, search through people they already know: an advisor's contacts, their lab, their department. That returns people who are already close. Yet new information travels through weak ties [5], and people who broker between groups that don't talk have better ideas [4]. At UIUC ([N] researchers, [M] units), two groups can work on the same problem for years without a joint paper. **BridgeFinder** finds these gaps in the co-authorship graph and names the people on the other side.

2 RELATED WORK

Illinois Experts [1] and Google Scholar search researchers by keyword but ignore network position: your co-author's co-author ranks like a stranger. LinkedIn's "People You May Know" and Twitter's Who-to-Follow [6] use triadic closure, the opposite of brokerage. Link prediction [7] finds *likely* ties; we rank ties by *value*, using Burt's constraint [3].

3 COURSE CONCEPTS & NETWORK ANALYSIS

Core concept: **structural holes and brokerage** (Lesson 1), with **community discovery** (Lesson 10) and **weak ties and decentralized search** (Lessons 1, 13).

Graph and method. Nodes are UIUC researchers in OpenAlex [9]. Edges are 2015–2024 co-authorships weighted by joint papers. Each node has a topic vector. (1) Louvain modularity [2] finds communities for the map. (2) Two communities form a *hole* when their topics are similar but few edges join them. (3) For a user u , each researcher v is scored by $\text{topic similarity} \times \text{graph distance} \times \text{brokerage gain}$, the drop in u 's constraint if the tie $u-v$ were added. (4) A weak-tie shortest path offers an introduction.

Alternatives from class. Triadic closure only finds people already near you; topic similarity alone can't tell a stranger from a friend of a friend. Girvan–Newman [8] could not split a 150-node graph in 148 s in our pilot; Louvain split 610 nodes in 1.5 s.

Claim that could be false. From the 2015–19 graph, BridgeFinder ranks the actual partners of new *cross-community* collaborations

(2020–24) higher than triadic-closure or topic-only ranking, and those collaborations' papers are cited more than new within-community ones in the same fields (Burt's claim).

4 PROPOSED WORK

A web app for Figure 1 (Python/networkx over an OpenAlex snapshot) plus the backtest. **Oct 30:** graph, holes, backtest, prototype, feedback round 1. **Mid-Nov.:** person cards, intro paths, round 2. **Dec 9:** round 3, evaluation. *Assumptions:* OpenAlex disambiguation is good enough (we will hand-check 50 authors); topic vectors capture "similar research"; students without papers are placed through their advisor or lab; using only public publication data keeps suggestions appropriate.

5 PRELIMINARY DESIGN

In Figure 1, the map comes first, so users see *where* the holes are before seeing names. Every suggestion shows its reasons, because unexplained recommendations are hard to trust.

6 USERS

UIUC students who want research mentors or collaborators. We will recruit 10–15 per round from our own networks and classes, deliberately spanning several majors so suggestions cross fields. Friends may be generous raters, so all ratings are blind. The same people test each revised version over three rounds.

7 EVALUATION

(1) **Backtest (no users).** Using only 2015–19 data, how often are the real partners of new cross-community collaborations (2020–24) in our top k , vs. triadic closure, topic-only and degree? (2) **Blind rating.** Each participant rates a mixed, unlabeled list of suggestions from BridgeFinder and the baselines as relevant and/or new to them. (3) **Usefulness:** would they want to contact each person? No one has to send an email. We succeed if we beat triadic closure on (1) and (2), and will say where we do not.

Use of AI: Claude (Anthropic) helped brainstorm, prototype and draft; we checked everything.

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